

3.6 Transformer on No Load

Consider a transformer on no load as shown in fig 3.3 (i). When the supply voltage, V_1 is applied to the primary, a current I_0 (no load current) will circulate through the primary winding. The primary no load current consists of two components.

- (i) a reactive or magnetising component I_m and
- (ii) an active or working component I_w .

The function of magnetising component of the no load current (I_m) is to produce magnetising flux and thus it will be in phase with the flux. Active component (I_w) produces the required real power to supply the hysteresis and eddy current losses in the iron core and the negligible copper losses in the primary winding. This component of no load current is in phase with the applied voltage V_1 . Fig. 3.3(ii) shows the phasor diagram of the transformer at no load condition. To draw the phasor diagram, take magnetic flux, ϕ as reference because, the flux, ϕ common to both the windings of the transformer. It is clear that the induced emf in the primary and secondary windings lag the flux by 90° . Consequently, emfs, E_1 and E_2 are drawn so that these emfs lag the flux ϕ by 90° .

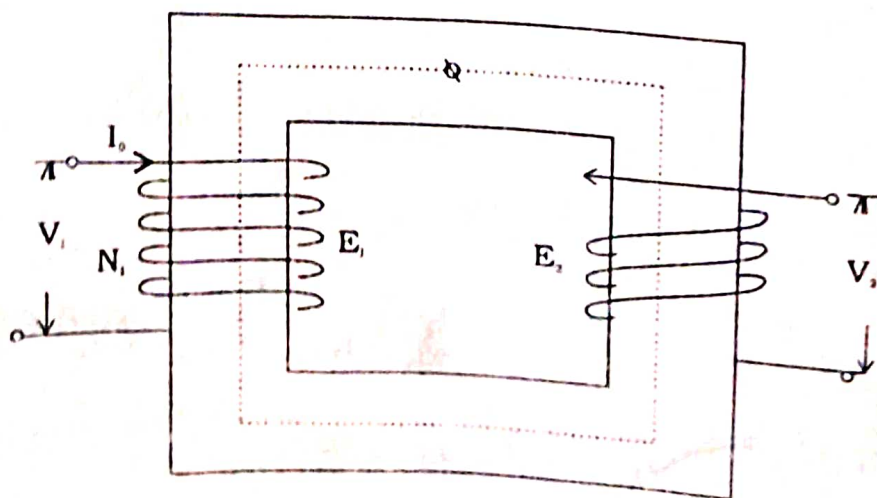


Fig. 3.3(a) Transformer at no load

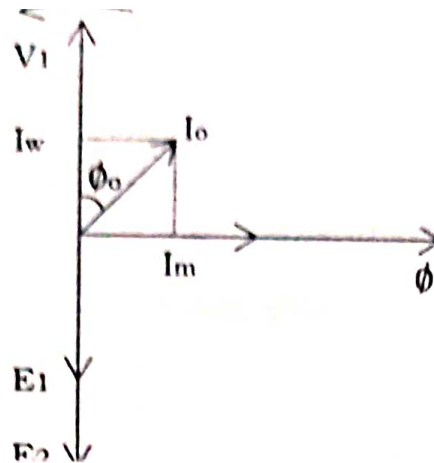


Fig. 3.3(b) *Phasor Diagram at no load*

Magnetising current, I_m can be drawn in phase with the flux. The phasor representing the applied voltage V_1 , can be drawn equal and opposite to the induced emf E_1 .

The active or working component (I_w) can be drawn in phase with the applied voltage V_1 . Then the phasor sum of I_w and I_m gives the no load primary current (I_0) as shown in figure 3.3 (ii).

From the phasor diagram,

$$I_w = I_0 \cos \phi$$

$$I_m = I_0 \sin \phi$$

$$\text{and } I_0 = \sqrt{I_w^2 + I_m^2}$$

As the magnitude of I_w is quite small compared to I_0 , the power factor of the transformer at no load is also very small.